HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Researched and prototyped novel mobile graphics algorithms for performance optimization in games, addressing identified scalability issues for partner studios.
- Designed and implemented computer graphics algorithms for mobile games, enhancing performance and extending battery life on client devices.
- Analyzed key developments in mobile computer graphics research from industry conferences (e.g., SIGGRAPH, Eurographics) to propose solutions for game engine graphics challenges.
- Developed C++ code for graphics rendering pipelines, focusing on optimization for mobile GPUs and CPUs to improve frame rates and reduce power consumption.
- Integrated and tested graphics algorithms within **Unity** and **Unreal Engine** environments, utilizing debugging tools like **RenderDoc** for performance analysis.
- Collaborated with international research teams and product groups to translate academic research into deployable product features and address partner technical needs.
- Containerized graphics rendering components using **Docker** and deployed them to a **Linux** environment, ensuring consistent performance across development and staging.

SKILLS

Graphics Fundamentals – Shading, Rasterization, Culling, Mobile GPU Optimization, Mobile CPU Optimization

Graphics APIs & Languages – HLSL, GLSL, Vulkan, GLES3

Programming Languages – C++, Python

Game Engines & Tools – Unity, Unreal Engine 4/5, Godot, RenderDoc, Pix, NVIDIA NSight

Development Methodologies – Agile, Scrum, AI agent literacy, Data-oriented design, ECS

PROJECTS

Mobile Game Rendering Performance Enhancements

- Developed and optimized **C++** rendering algorithms for a mobile game prototype, focusing on reducing draw calls and improving shader efficiency.
- Integrated **Unity** engine features and utilized **RenderDoc** to profile GPU performance, identifying and resolving bottlenecks that impacted frame rate by **~18%**.
- Contributed to a shared library of graphics shaders written in **GLSL** for cross-project use, improving code reusability across mobile titles.

Cross-Platform Graphics Shader Optimization

- Engineered a set of reusable graphics shaders in **HLSL** and **GLSL** for common rendering effects like deferred lighting and post-processing.
- Refactored existing shader code to improve performance on mobile GPUs, achieving a reduction in shader compilation time by **~30%** and GPU execution time by **15%**.
- Implemented a testing harness using **Python** to automate shader compilation and basic visual validation across different target platforms.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)